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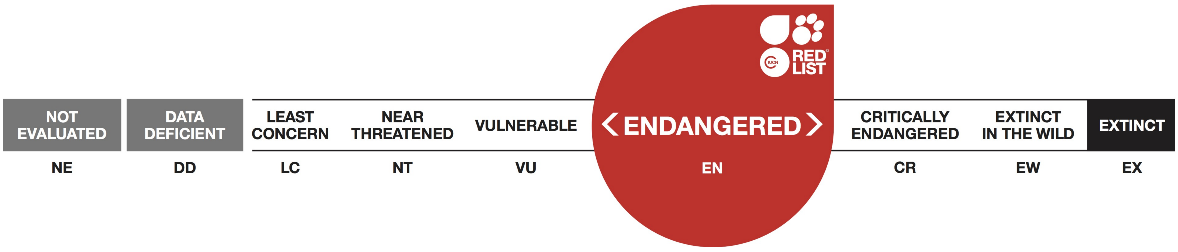
Acheilognathus longipinnis, Deepbodied Bitterling

Assessment by: Hasegawa, K., Kanao, S., Miyazaki, Y., Mukai, T., Nakajima, J., Takaku, K. & Taniguchi, Y.



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If you see any errors or have any questions or suggestions on what is shown in this document, please provide us with feedback so that we can correct or extend the information provided.

Taxonomy

Kingdom	Phylum	Class	Order	Family
Animalia	Chordata	Actinopterygii	Cypriniformes	Cyprinidae

Taxon Name: *Acheilognathus longipinnis* Regan, 1905

Common Name(s):

- English: Deepbodied Bitterling, Itasenpara Bitterling

Assessment Information

Red List Category & Criteria: Endangered B2ab(ii,iii,v) ver 3.1

Year Published: 2019

Date Assessed: December 7, 2017

Justification:

This species is endemic to Japan where it occurs in four rivers in four prefectures (Toyama, Gifu, Aichi and Osaka). The extent of occurrence (EOO) is 9,060 km^2 and it has an area of occupancy (AOO) of 36 km^2. There are three locations based on the primary threats of land reclamation, dam construction and river channelisation. These threats are leading to continuing declines in the AOO, habitat and population size. Therefore, this species is assessed as Endangered.

Previously Published Red List Assessments

1996 – Vulnerable (VU)
http://dx.doi.org/10.2305/IUCN.UK.1996.RLTS.T213A13052654.en

1994 – Endangered (E)

Geographic Range

Range Description:

This species is endemic to Japan where it occurs in four prefectures (Toyama, Gifu, Aichi and Osaka). This species has been recorded from Busshoji, Kiso, Yodo and Mo-o Rivers (K. Watanabe pers. comm. 2017).

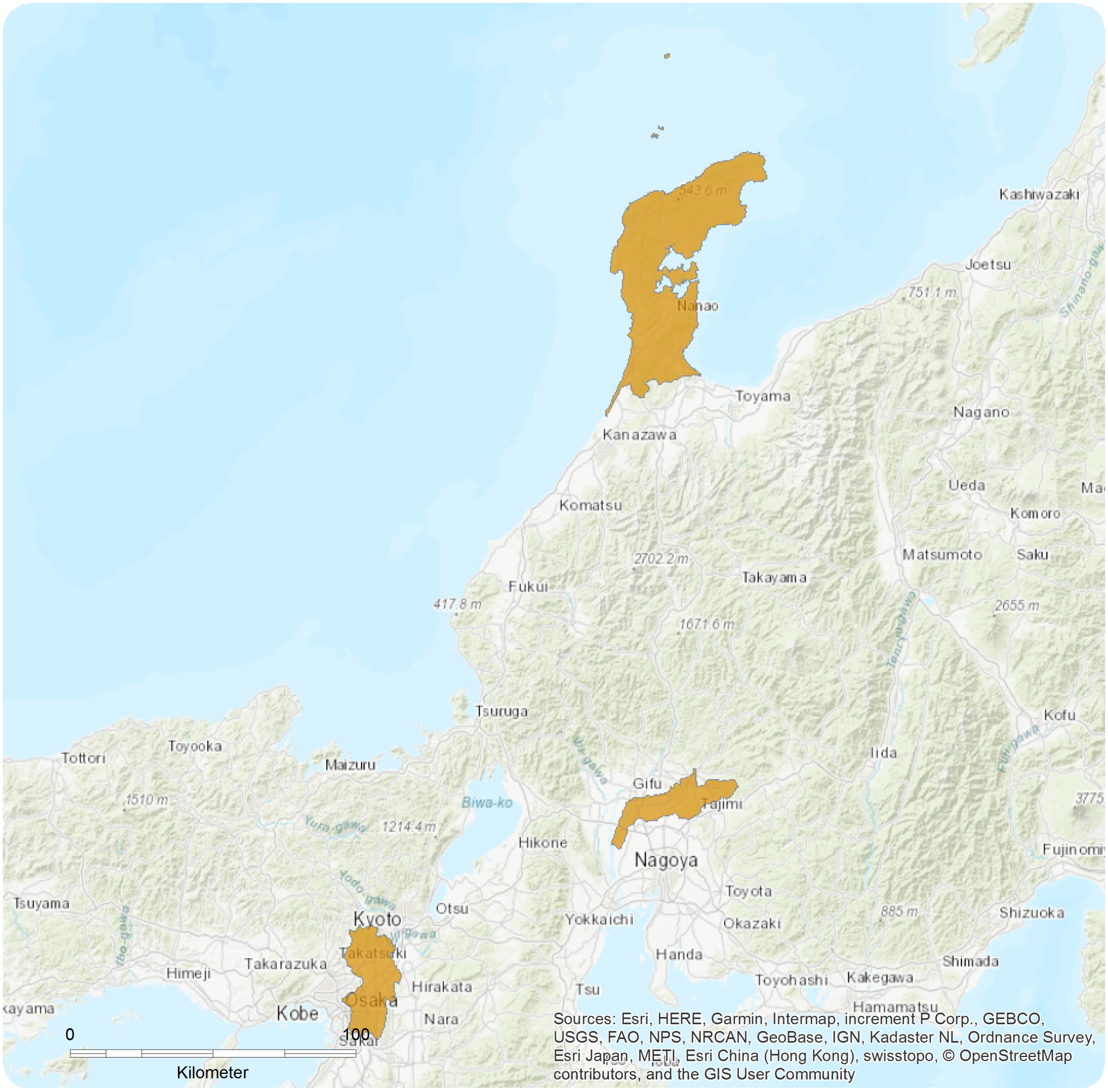
The extent of occurrence (EOO) is 9,060 km^2 and it has an area of occupancy (AOO) of 36 km^2. There are three locations: 1) Busshoji and Mo-o Rivers based on the threat of land reclamation; 2) Kiso River based on the threat of dam construction and river channelisation; and 3) Yodo River, also based on the threat of dam construction and river channelisation.

Country Occurrence:

Native: Japan (Honshu)

Distribution Map

Acheilognathus longipinnis

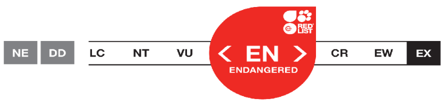


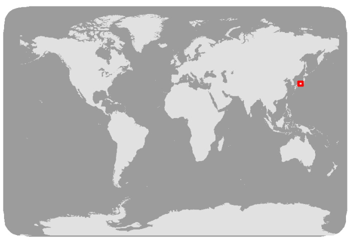
Range

Extant (resident)

Compiled by:

IUCN (International Union for Conservation of Nature)





The boundaries and names shown and the designations used on this map do not imply any official endorsement, acceptance or opinion by IUCN.

Population

In the Busshoji River (Toyama Prefecture) no mature individuals have been recorded for the past 10 years. However, in 2010 and 2016, 25 and two juveniles, respectively, were recorded.

In Mo-o River (Toyama Prefecture) 2,301 juveniles were recorded from the main body and 134 juveniles were recorded from its tributaries, in 2010-2011.

In the Yodo River (Osaka Prefucture) 500 mature individuals were introduced (from a captive bred population) at one locality in 2013. Prior to this stocking event, no juveniles had been recorded at the locality for seven years. After this stocking event several hundred juveniles have been recorded at the locality in subsequent years. At a second locality in the river, 500 mature individuals were introduced (from a captive bred population) in 2009 and 2011. Prior to the 2009 stocking event, no juveniles had been recorded at the locality since at least 2001. Numbers of juveniles increased after the stocking event to a peak in 2013 but decreased to zero in 2015 and 2016 (K. Watanabe pers. comm. 2017).

Current Population Trend: Decreasing

Habitat and Ecology (see Appendix for additional information)

This species inhabits stagnant or slow running waters in rivers, ponds and creeks (Kawamura 2014).

Systems: Freshwater

Use and Trade

This species is harvested for ornamental use and is displayed within Japan (Kawamura 2014).

Threats (see Appendix for additional information)

The primary threats to this species are land reclamation, construction of dams and river channelisation, pollution, competition with the introduced *Rhodeus ocellatus* ssp. *ocellatus*, predation by the introduced Largemouth Bass (*Micropterus salmoides*) and Bluegill (*Lepomis macrochirus* ssp. *macrochirus*) and harvesting for ornamental use (Kawamura 2014). This species lays its eggs within freshwater mussels (*Unio*). These mussels are susceptible to water pollution and the decline in this fish species can be linked to the decline in the mussel (S. Kanao and J. Nakajima pers. comm. 2017).

Conservation Actions (see Appendix for additional information)

Actions in place include stocking of mature individuals from *ex situ* conservation projects (propagation of fishes for reintroduction into the rivers), removal of exotic species, habitat restoration and education/awareness raising of the public (Ogawa 2008). This species has been designated as a national monument in Japan under the Act on Protection of Cultural Properties.

Credits

Assessor(s): Hasegawa, K., Kanao, S., Miyazaki, Y., Mukai, T., Nakajima, J., Takaku, K. & Taniguchi, Y.

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Facilitators(s) and Compiler(s):	Sayer, C.

Bibliography

IUCN. 2019. The IUCN Red List of Threatened Species. Version 2019-2. Available at: www.iucnredlist.org. (Accessed: 04 July 2019).

Kawamura, K. 2015. *Acheilognathus longipinnis*. *Threatened Wildlife of Japan Red Data Book 2014; Pisces - Brackish and Fresh Water Fishes*, pp. 20-21. Gyosei, Tokyo.

Ogawa, R. 2008. *Acheilognathus longipinnis*: a symbol fish of flood plains with natural hydrometeorological environments. *Japanese Journal of Ichthyology* 55: 144-148.

Citation

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External Resources

For Images and External Links to Additional Information, please see the Red List website.

Appendix

Habitats

(http://www.iucnredlist.org/technical-documents/classification-schemes)

Habitat	Season	Suitability	Major Importance?
5. Wetlands (inland) -> 5.1. Wetlands (inland) - Permanent Rivers/Streams/Creeks (includes waterfalls)	-	Suitable	-

Threats

(http://www.iucnredlist.org/technical-documents/classification-schemes)

Threat	Timing	Scope	Severity	Impact Score
5. Biological resource use -> 5.4. Fishing & harvesting aquatic resources -> 5.4.1. Intentional use: (subsistence/small scale) [harvest]	Ongoing	Whole (>90%)	Slow, significant declines	Medium impact: 7
	Stresses:	2. Species Stresses -> 2.1. Species mortality		
7. Natural system modifications -> 7.2. Dams & water management/use -> 7.2.10. Large dams	Ongoing	Majority (50-90%)	Slow, significant declines	Medium impact: 6
	Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation		
7. Natural system modifications -> 7.3. Other ecosystem modifications	Ongoing	Whole (>90%)	Slow, significant declines	Medium impact: 7
	Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation		
8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.2. Named species (Rhodeus ocellatus ssp. ocellatus)	Ongoing	Whole (>90%)	Slow, significant declines	Medium impact: 7
	Stresses:	2. Species Stresses -> 2.3. Indirect species effects -> 2.3.2. Competition		
8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.2. Named species (Lepomis macrochirus)	Ongoing	Whole (>90%)	Rapid declines	High impact: 8
	Stresses:	2. Species Stresses -> 2.1. Species mortality		
8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.2. Named species (Micropterus salmoides)	Ongoing	Whole (>90%)	Rapid declines	High impact: 8
	Stresses:	2. Species Stresses -> 2.1. Species mortality		
9. Pollution -> 9.1. Domestic & urban waste water -> 9.1.1. Sewage	Ongoing	Whole (>90%)	Slow, significant declines	Medium impact: 7
	Stresses:	1. Ecosystem stresses -> 1.2. Ecosystem degradation 2. Species Stresses -> 2.2. Species disturbance 2. Species Stresses -> 2.3. Indirect species effects -> 2.3.3. Loss of mutualism		

9. Pollution -> 9.1. Domestic & urban waste water -> 9.1.2. Run-off	Ongoing	Whole (>90%)	Slow, significant declines	Medium impact: 7
	Stresses:	1. Ecosystem stresses -> 1.2. Ecosystem degradation 2. Species Stresses -> 2.2. Species disturbance 2. Species Stresses -> 2.3. Indirect species effects -> 2.3.3. Loss of mutualism		
9. Pollution -> 9.3. Agricultural & forestry effluents -> 9.3.1. Nutrient loads	Ongoing	Whole (>90%)	Slow, significant declines	Medium impact: 7
	Stresses:	1. Ecosystem stresses -> 1.2. Ecosystem degradation 2. Species Stresses -> 2.2. Species disturbance 2. Species Stresses -> 2.3. Indirect species effects -> 2.3.3. Loss of mutualism		

Conservation Actions in Place
(http://www.iucnredlist.org/technical-documents/classification-schemes)

Conservation Actions in Place
In-Place Land/Water Protection and Management
Invasive species control or prevention: Yes
In-Place Species Management
Subject to ex-situ conservation: Yes

Conservation Actions Needed
(http://www.iucnredlist.org/technical-documents/classification-schemes)

Conservation Actions Needed
1. Land/water protection -> 1.1. Site/area protection
1. Land/water protection -> 1.2. Resource & habitat protection
2. Land/water management -> 2.1. Site/area management
2. Land/water management -> 2.2. Invasive/problematic species control
2. Land/water management -> 2.3. Habitat & natural process restoration
3. Species management -> 3.2. Species recovery
3. Species management -> 3.3. Species re-introduction -> 3.3.1. Reintroduction
3. Species management -> 3.4. Ex-situ conservation -> 3.4.1. Captive breeding/artificial propagation
4. Education & awareness -> 4.3. Awareness & communications
5. Law & policy -> 5.1. Legislation -> 5.1.2. National level

Research Needed
(http://www.iucnredlist.org/technical-documents/classification-schemes)

Research Needed
1. Research -> 1.2. Population size, distribution & trends
1. Research -> 1.3. Life history & ecology
1. Research -> 1.5. Threats
1. Research -> 1.6. Actions
2. Conservation Planning -> 2.1. Species Action/Recovery Plan
2. Conservation Planning -> 2.2. Area-based Management Plan
3. Monitoring -> 3.1. Population trends

Additional Data Fields

Distribution
Estimated area of occupancy (AOO) (km^2): 36
Continuing decline in area of occupancy (AOO): Yes
Estimated extent of occurrence (EOO) (km^2): 9060
Number of Locations: 3
Population
Continuing decline of mature individuals: Yes
Habitats and Ecology
Continuing decline in area, extent and/or quality of habitat: Yes

The IUCN Red List Partnership

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